Derivation of the Compressible Euler Equations from the Boltzmann Equation

1 The Boltzmann Equation

We begin with the Boltzmann equation without external acceleration:

$$\frac{\partial f}{\partial t} + v \cdot \nabla_x f = Q(f, f). \tag{1}$$

Here

- $f = f(x, v, t)$ is the molecular distribution function,
- $x \in \mathbb{R}^3$ is position,
- $v \in \mathbb{R}^3$ is molecular velocity,
- $Q(f, f)$ is the collision operator.

The quantity

$$f(x, v, t)\, dx\, dv \tag{2}$$

represents the number of molecules near position x, with velocity near v, at time t.

2 Macroscopic Variables

Let m_0 be the mass of one molecule. We define the mass density by

$$\rho = m_0 \int_{\mathbb{R}^3} f\, dv. \tag{3}$$

The macroscopic velocity u is the average molecular velocity:

$$\rho u = m_0 \int_{\mathbb{R}^3} v f\, dv. \tag{4}$$

Define the peculiar velocity

$$c = v - u. \tag{5}$$

Thus

$$v = u + c. \tag{6}$$

By definition of u,

$$m_0 \int_{\mathbb{R}^3} c f\, dv = 0. \tag{7}$$

Indeed,

$$m_0 \int cf \, dv = m_0 \int (v-u) f \, dv \tag{8}$$

$$= m_0 \int vf \, dv - u \, m_0 \int f \, dv \tag{9}$$

$$= \rho u - \rho u \tag{10}$$

$$= 0. \tag{11}$$

The internal energy per unit mass is defined by

$$\rho e = \frac{m_0}{2} \int_{\mathbb{R}^3} |c|^2 f \, dv. \tag{12}$$

The total energy density is

$$E = \rho e + \frac{1}{2} \rho |u|^2. \tag{13}$$

3 Collision Invariants

The collision operator conserves mass, momentum, and kinetic energy. Therefore,

$$\int_{\mathbb{R}^3} Q(f,f) \, dv = 0, \tag{14}$$

$$\int_{\mathbb{R}^3} v Q(f,f) \, dv = 0, \tag{15}$$

and

$$\int_{\mathbb{R}^3} |v|^2 Q(f,f) \, dv = 0. \tag{16}$$

These identities are exact. They express the fact that binary collisions redistribute molecular velocities, but do not create or destroy total mass, momentum, or energy.

4 Mass Equation

Integrate the Boltzmann equation over velocity:

$$\int \frac{\partial f}{\partial t} \, dv + \int v \cdot \nabla_x f \, dv = \int Q(f,f) \, dv. \tag{17}$$

Using the first collision invariant,

$$\int Q(f,f) \, dv = 0. \tag{18}$$

Therefore,

$$\frac{\partial}{\partial t} \int f \, dv + \nabla_x \cdot \int vf \, dv = 0. \tag{19}$$

Multiplying by m_0, we obtain

$$\frac{\partial}{\partial t} \left(m_0 \int f \, dv \right) + \nabla_x \cdot \left(m_0 \int vf \, dv \right) = 0. \tag{20}$$

Using

$$\rho = m_0 \int f \, dv, \qquad \rho u = m_0 \int vf \, dv, \tag{21}$$

we obtain the mass equation:

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla_x \cdot (\rho u) = 0.} \tag{22}$$

5 Momentum Equation

Multiply the Boltzmann equation by $m_0 v$ and integrate over velocity:

$$\frac{\partial}{\partial t}\left(m_0 \int v f \, dv\right) + \nabla_x \cdot \left(m_0 \int v \otimes v \, f \, dv\right) = m_0 \int v Q(f, f) \, dv. \tag{23}$$

Using momentum conservation in collisions,

$$m_0 \int v Q(f, f) \, dv = 0. \tag{24}$$

Thus

$$\frac{\partial(\rho u)}{\partial t} + \nabla_x \cdot \left(m_0 \int v \otimes v \, f \, dv\right) = 0. \tag{25}$$

Now use

$$v = u + c. \tag{26}$$

Then

$$v \otimes v = (u + c) \otimes (u + c). \tag{27}$$

Expanding,

$$v \otimes v = u \otimes u + u \otimes c + c \otimes u + c \otimes c. \tag{28}$$

Therefore,

$$m_0 \int v \otimes v \, f \, dv = m_0 \int u \otimes u \, f \, dv + m_0 \int u \otimes c \, f \, dv \tag{29}$$

$$+ m_0 \int c \otimes u \, f \, dv + m_0 \int c \otimes c \, f \, dv. \tag{30}$$

Since u does not depend on v,

$$m_0 \int u \otimes u \, f \, dv = \rho u \otimes u. \tag{31}$$

Also,

$$m_0 \int c f \, dv = 0, \tag{32}$$

so the mixed terms vanish:

$$m_0 \int u \otimes c \, f \, dv = 0, \qquad m_0 \int c \otimes u \, f \, dv = 0. \tag{33}$$

Define the pressure tensor

$$P = m_0 \int c \otimes c \, f \, dv. \tag{34}$$

Then

$$m_0 \int v \otimes v \, f \, dv = \rho u \otimes u + P. \tag{35}$$

Thus the exact momentum equation is

$$\boxed{\frac{\partial(\rho u)}{\partial t} + \nabla_x \cdot (\rho u \otimes u + P) = 0.} \tag{36}$$

6 Energy Equation

Multiply the Boltzmann equation by the molecular kinetic energy

$$\frac{1}{2} m_0 |v|^2 \tag{37}$$

and integrate over velocity:

$$\frac{\partial}{\partial t}\left(\int \frac{1}{2} m_0 |v|^2 f\, dv\right) + \nabla_x \cdot \left(\int \frac{1}{2} m_0 |v|^2 v f\, dv\right) = \int \frac{1}{2} m_0 |v|^2 Q(f,f)\, dv. \tag{38}$$

Using energy conservation in collisions,

$$\int \frac{1}{2} m_0 |v|^2 Q(f,f)\, dv = 0. \tag{39}$$

Define total energy density:

$$E = \int \frac{1}{2} m_0 |v|^2 f\, dv. \tag{40}$$

Therefore,

$$\frac{\partial E}{\partial t} + \nabla_x \cdot \left(\int \frac{1}{2} m_0 |v|^2 v f\, dv\right) = 0. \tag{41}$$

Now decompose the energy density using $v = u + c$:

$$|v|^2 = |u + c|^2 = |u|^2 + 2u \cdot c + |c|^2. \tag{42}$$

Hence

$$E = \frac{1}{2} m_0 \int |v|^2 f\, dv \tag{43}$$

$$= \frac{1}{2} |u|^2 m_0 \int f\, dv + u \cdot m_0 \int c f\, dv + \frac{1}{2} m_0 \int |c|^2 f\, dv. \tag{44}$$

The middle term vanishes. Thus

$$E = \frac{1}{2} \rho |u|^2 + \rho e. \tag{45}$$

The exact energy flux is

$$\mathcal{F}_E = \int \frac{1}{2} m_0 |v|^2 v f\, dv. \tag{46}$$

Using $v = u + c$, this flux decomposes as

$$\mathcal{F}_E = E u + P u + q, \tag{47}$$

where

$$q = \frac{1}{2} m_0 \int |c|^2 c f\, dv \tag{48}$$

is the heat flux.

Therefore, the exact energy equation is

$$\boxed{\frac{\partial E}{\partial t} + \nabla_x \cdot (E u + P u + q) = 0.} \tag{49}$$

7 What Is Exact and What Requires Approximation?

Up to this point, the derivation is exact, assuming only the validity of the Boltzmann equation and the conservation properties of the collision operator.

The exact moment system is

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla_x \cdot (\rho u) = 0,} \tag{50}$$

$$\boxed{\frac{\partial (\rho u)}{\partial t} + \nabla_x \cdot (\rho u \otimes u + P) = 0,} \tag{51}$$

$$\boxed{\frac{\partial E}{\partial t} + \nabla_x \cdot (E u + P u + q) = 0.} \tag{52}$$

Here

$$P = m_0 \int c \otimes c \, f \, dv \tag{53}$$

and

$$q = \frac{1}{2} m_0 \int |c|^2 c \, f \, dv. \tag{54}$$

This system is not closed because P and q depend on higher moments of f.

The first approximation needed to obtain the Euler equations is the local equilibrium approximation:

$$f \approx f_M, \tag{55}$$

where f_M is a local Maxwellian.

For a local Maxwellian, the random molecular motion is isotropic, and therefore

$$P = pI. \tag{56}$$

Also, the heat flux vanishes:

$$q = 0. \tag{57}$$

Thus,

$$Eu + Pu + q = Eu + pIu = (E + p)u. \tag{58}$$

This is the point at which the exact kinetic moment equations become the Euler equations.

8 Euler Equations

Using the Euler closure

$$P = pI, \qquad q = 0, \tag{59}$$

we obtain

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla_x \cdot (\rho u) = 0,} \tag{60}$$

$$\boxed{\frac{\partial (\rho u)}{\partial t} + \nabla_x \cdot (\rho u \otimes u + pI) = 0,} \tag{61}$$

$$\boxed{\frac{\partial E}{\partial t} + \nabla_x \cdot ((E + p)u) = 0.} \tag{62}$$

These are the compressible Euler equations in conservative form without external acceleration.

9 Equation of State

The Euler equations are not closed until one specifies an equation of state.

For a calorically perfect ideal gas,

$$p = \rho RT, \tag{63}$$

and

$$e = c_v T. \tag{64}$$

Thus

$$T = \frac{e}{c_v}, \tag{65}$$

so

$$p = \rho R \frac{e}{c_v}. \tag{66}$$

Since

$$\gamma = \frac{c_p}{c_v}, \qquad R = c_p - c_v, \tag{67}$$

we have

$$\frac{R}{c_v} = \gamma - 1. \tag{68}$$

Therefore,

$$\boxed{p = (\gamma - 1)\rho e.} \tag{69}$$

Since

$$E = \rho e + \frac{1}{2}\rho|u|^2, \tag{70}$$

we have

$$\rho e = E - \frac{1}{2}\rho|u|^2. \tag{71}$$

Hence

$$\boxed{p = (\gamma - 1)\left(E - \frac{1}{2}\rho|u|^2\right).} \tag{72}$$

This equation of state is an additional thermodynamic closure assumption. It is not obtained from conservation laws alone.

10 Meaning of $E + p$

The total energy density is

$$E = \rho e + \frac{1}{2}\rho|u|^2. \tag{73}$$

The term $E + p$ appears because a moving fluid carries its own energy and also performs pressure work on neighboring fluid.

The energy flux in the Euler equations is

$$(E + p)u. \tag{74}$$

Equivalently, define the specific enthalpy

$$h = e + \frac{p}{\rho}. \tag{75}$$

Then

$$E + p = \rho h + \frac{1}{2}\rho|u|^2. \tag{76}$$

Thus $E + p$ is the total enthalpy density, including kinetic energy.

11 One-Dimensional Euler Equations in Eulerian Coordinates

In one spatial dimension, the Euler equations are

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} = 0, \tag{77}$$

$$\frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u^2 + p)}{\partial x} = 0, \tag{78}$$

$$\frac{\partial E}{\partial t} + \frac{\partial((E + p)u)}{\partial x} = 0. \tag{79}$$

Here

$$E = \rho e + \frac{1}{2}\rho u^2, \tag{80}$$

and for an ideal gas,

$$p = (\gamma - 1)\left(E - \frac{1}{2}\rho u^2\right). \tag{81}$$

12 One-Dimensional Euler Equations in Lagrangian Coordinates

Define the Lagrangian mass coordinate m by

$$m(x,t) = \int_{x_0}^{x} \rho(\xi, t)\, d\xi. \tag{82}$$

Then

$$\frac{\partial m}{\partial x} = \rho, \qquad \frac{\partial x}{\partial m} = \frac{1}{\rho}. \tag{83}$$

The coordinate m labels a fixed amount of mass. In Lagrangian coordinates, the independent variables are m and t.

The standard Lagrangian mass equation is

$$\frac{\partial}{\partial t}\left(\frac{1}{\rho}\right) - \frac{\partial u}{\partial m} = 0. \tag{84}$$

If we keep ρ instead of introducing the specific volume $1/\rho$, this is equivalently

$$\boxed{\frac{\partial \rho}{\partial t} + \rho^2 \frac{\partial u}{\partial m} = 0.} \tag{85}$$

The momentum equation is

$$\boxed{\frac{\partial u}{\partial t} + \frac{\partial p}{\partial m} = 0.} \tag{86}$$

The energy equation is

$$\boxed{\frac{\partial E}{\partial t} + \frac{\partial (pu)}{\partial m} = 0.} \tag{87}$$

Thus, using ρ rather than the specific volume, the one-dimensional Euler equations in Lagrangian mass coordinates are

$$\boxed{\frac{\partial \rho}{\partial t} + \rho^2 \frac{\partial u}{\partial m} = 0,} \tag{88}$$

$$\boxed{\frac{\partial u}{\partial t} + \frac{\partial p}{\partial m} = 0,} \tag{89}$$

$$\boxed{\frac{\partial E}{\partial t} + \frac{\partial (pu)}{\partial m} = 0.} \tag{90}$$

The ideal-gas closure is

$$\boxed{p = (\gamma - 1)\left(E - \frac{1}{2}\rho u^2\right).} \tag{91}$$

13 Summary

The derivation has the following structure:

$$\boxed{\text{Boltzmann equation} \longrightarrow \text{exact moment equations with } P \text{ and } q} \tag{92}$$

$$\boxed{\text{Euler approximation: } f \approx f_M, \quad P = pI, \quad q = 0} \tag{93}$$

$$\boxed{\text{EOS closure: } p = (\gamma - 1)\rho e} \tag{94}$$

Thus the exact part is the moment derivation up to P and q. The Euler equations require the local Maxwellian approximation, and the closed ideal-gas Euler system requires an equation of state.